FLYRANK GENERAL AI FLUENCY — CAPSTONE PROJECT (WEEK 10)

Send the Link: Launch, Demo & Story

Comprehensive 5-Page Capstone Synthesis Document | Student: ABDUL SAMI UTHWAL

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Track: General AI Fluency & Machine Learning

Live Launch URL: abdulsamiuthwal-portfolio.vercel.app

Live Demo Video: youtu.be/-uCqi_JDjfY

GitHub Repository: abdulsamiuthwal-eng/Flyrank-assignments

Completion Date: 2026-08-12

1. Executive Summary & Capstone Vision

A portfolio is not a static PDF or classroom assignment—it is a live, verifiable career platform that proves what you can build, deploy, and defend. This Capstone deliverable synthesises the three core pillars of the FlyRank General AI Fluency Track:

- **Pillar 1 — The Launch:** Public deployment of a claim-first AI systems portfolio over HTTPS on Vercel Edge Network, fortified with Vercel Web Analytics, OpenGraph social cards, and the FlyRank Verified Graduate Badge.
- **Pillar 2 — The Demo:** Live video proof (`https://youtu.be/-uCqi_JDjfY`) demonstrating autonomous AI agent execution, API response latency, and verbal explanations of design trade-offs and limitations on camera.
- **Pillar 3 — The Story:** The complete 10-week engineering journey from CTR opportunity scoring to champion gradient boosting models (F1: 0.783), autonomous research agents, serverless lead capture pipelines, and production site hardening.

2. Pillar 1: The Launch Infrastructure

Live Production Environment Details

- **Public Production Domain:** `https://abdulsamiuthwal-portfolio.vercel.app/`
- **Security & Protocol:** HTTPS SSL/TLS Certificate auto-provisioned via Vercel Edge Network.
- **Analytics Engine:** Vercel Insights & Speed Analytics script (`/_vercel/insights/script.js`) active in `<head>`.
- **Launch Hygiene Verification:**
  - Custom SVG Lightning Favicon (`data:image/svg+xml...`)
  - OpenGraph Cards (`og:title`, `og:description`, `og:image`, `og:url`)
  - Twitter Summary Card (`summary_large_image`)
  - Schema.org `Person` JSON-LD structured data
- **FlyRank Credential Badge:** Embedded FlyRank Verified Graduate Badge in footer linking to `https://internship.flyrank.ai`.

3. Pillar 2: The Demo — Autonomous Agent & Live Video Proof

Live Video Proof Metadata

YouTube Unlisted Video Link: https://youtu.be/-uCqi_JDjfY

Duration & Narration: Live end-to-end screen recording with voice narration demonstrating the `ResearchScout` AI Agent and Web UI.

Autonomous Agent Architecture (`ResearchScoutAgent`)

The `ResearchScout` agent is an autonomous literature research assistant built in Python that executes a 4-step pipeline:

```
User Query (arXiv ID: 2312.00752)
       │
       ▼
┌──────────────────────────────────┐
│ TOOL 1 — arXiv REST API Fetch    │  ← Live HTTP GET to export.arxiv.org
└──────────────────────────────────┘
                │ Atom XML Payload
                ▼
┌──────────────────────────────────┐
│ XML Parser Engine                │  ← xml.etree.ElementTree
└──────────────────────────────────┘
                │ Metadata Dict
                ▼
┌──────────────────────────────────┐
│ CORE JOB — Synthesis Engine      │  ← 5-Section Markdown Brief
└──────────────────────────────────┘
                │ Markdown String
                ▼
┌──────────────────────────────────┐
│ TOOL 2 — Local Filesystem Storage│  ← Persists to work/outputs/research_briefs/
└──────────────────────────────────┘
```

Verbal Explanations On Camera

Requirement	Topic Explained On Camera	Engineering Rationale / Insight
1. Design Decision	Atom XML API vs. Full PDF Parsing	Switched from `pdfplumber` PDF parsing to arXiv Atom XML API feed. Reduced latency from **18 seconds → 1.4 seconds** (a **12× speedup**) while preserving full title, author, and abstract fidelity.
2. Known Limitation	Template-Based Synthesis Sections	Sections 2–4 (Architecture, Benchmarks, Gotchas) use a fixed engineering template derived from the abstract. Dynamic body extraction is planned for v3.

4. Pillar 3: The Story — 10-Week Engineering Narrative

The portfolio tells the progressive story of an AI engineer mastering machine learning, autonomous systems, serverless pipelines, and production deployment.

Milestone Breakdown Across 10 Weeks

Phase 1: Signal Framing & Data Contract (Weeks 1–4)
Engineered the search ranking CTR Opportunity Scoring signal (`expected_ctr - actual_ctr`) across 600 synthetic search landing pages and 15 domain clusters. Formulated data contracts and schemas for historical impressions and rankings.

Phase 2: ML Model Benchmarking & GroupKFold Validation Audit (Weeks 5–6)
Benchmarked 5 candidate machine learning models. Evaluated under strict GroupKFold domain-split validation to eliminate data leakage across domain clusters.

Model Candidate	Validation Strategy	Validated F1 Score	ROC-AUC Score
Baseline Rule Model	Heuristic Threshold	0.480	N/A
Logistic Regression	GroupKFold (k=5)	0.273	0.612
Decision Tree Classifier	GroupKFold (k=5)	0.583	0.741
Random Forest Classifier	GroupKFold (k=5)	0.609	0.812
Gradient Boosting (Champion)	GroupKFold (k=5)	**0.783**	**0.983**

Phase 3: Autonomous Agent & Live Lead Capture Pipeline (Weeks 7–8)
Developed the `ResearchScout` Python agent and built an end-to-end live Lead Capture System wired to Web3Forms serverless API on the production portfolio, delivering leads directly to Gmail.

5. Site Hardening, QA Stress Matrix & Performance Audit

In Week 9 ("Break Your Own Site"), the application was subjected to rigorous stress testing rather than relying on happy-path demonstrations.

"Where It Breaks" Stress Matrix

Scenario	Trigger Action	Initial Behavior	Triage	Remediation Applied
1. Empty Form	Clicked Submit with empty inputs	Native browser alert	Fix-Now	Added input trimming, regex email validation, and inline `#form-status` feedback.
2. Double Click	Clicked Submit twice in <300ms	Duplicate HTTP requests	Fix-Now	Added `isSubmitting` locking flag & button disabling.
3. Whitespace	Entered spaces " "	Accepted as valid text	Fix-Now	Enforced `.trim()` sanitization across all input fields.
4. Social Meta	Link shared on social media	Blank link preview	Fix-Now	Injected OpenGraph tags, Twitter Cards, & Schema JSON-LD.
5. API Rate Limit	>3 API requests / sec	HTTP 429 throttling	Limitation	Documented informal arXiv API rate limits with retry status.

Performance & PageSpeed Audit Scores

Audit Metric	Measurement Tool	Score / Result	Status
Page Speed Score	Lighthouse / PageSpeed	**98 / 100**	PASS
First Contentful Paint (FCP)	Navigation Timing API	**0.4 seconds**	PASS
Largest Contentful Paint (LCP)	Performance Observer	**0.8 seconds**	PASS
Cumulative Layout Shift (CLS)	Layout Instability API	**0.00**	PASS
Agent REST API Latency	Flask Server / arXiv	**681ms - 898ms**	PASS

6. Growth Roadmap & Preserved AI Context

The 3-Beat Case Study Addition Protocol

To ensure long-term portfolio growth without CSS rebuilds, every future case study follows a strict 3-beat structure:

- **Beat 1 (Problem):** Define the operational bottleneck and baseline delay.
- **Beat 2 (What You Did):** Detail the AI architecture, models, vector indexes, and agent loops deployed.
- **Beat 3 (What Came of It):** State the quantifiable ROI (% latency drop, accuracy score, hours saved).

Named Next Real Piece of Work

Project Name: *Autonomous Multi-Agent Enterprise RAG & Legal Citation Engine*

Architecture: Qdrant vector index + OpenAI `text-embedding-3-large` + Dual-agent supervisor loop (`LegalResearcher` + `CitationAuditor`) deployed on Cloud Run.

Target Metric: 85% reduction in legal document audit time with 99.2% citation precision.

7. Final Verification Checklist & Data Credit

Verification Item	Requirement Confirmation	Status
Live HTTPS Launch	Live at `https://abdulsamiuthwal-portfolio.vercel.app/` over SSL	VERIFIED
Live Demo Video	Unlisted YouTube video `https://youtu.be/-uCqi_JDjfY`	VERIFIED
Research Paper Deployed	Live at `https://abdulsamiuthwal-portfolio.vercel.app/work/paper/index.html`	VERIFIED
Graduate Credential Badge	Embedded in footer linking to `https://internship.flyrank.ai`	VERIFIED
Data Credit & Acknowledgment	"Built on the FlyRank ML Internship dataset" linking to flyrank.ai	VERIFIED

⚡ **Capstone Completion Confirmed — Abdul Sami Uthwal | FlyRank Internship 2026**